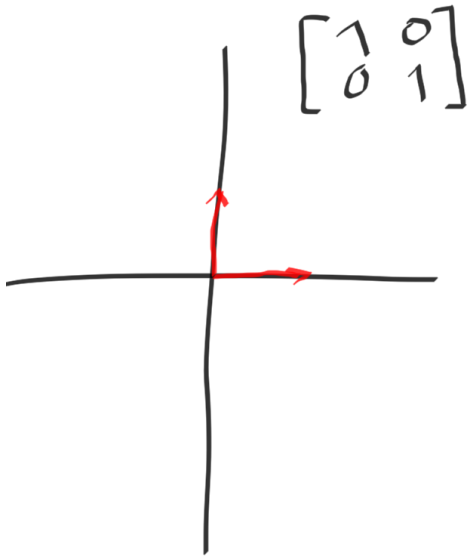
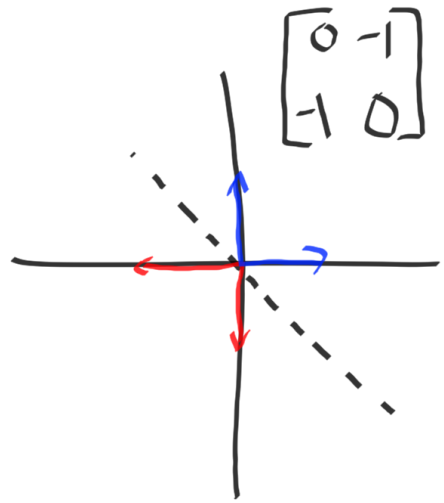
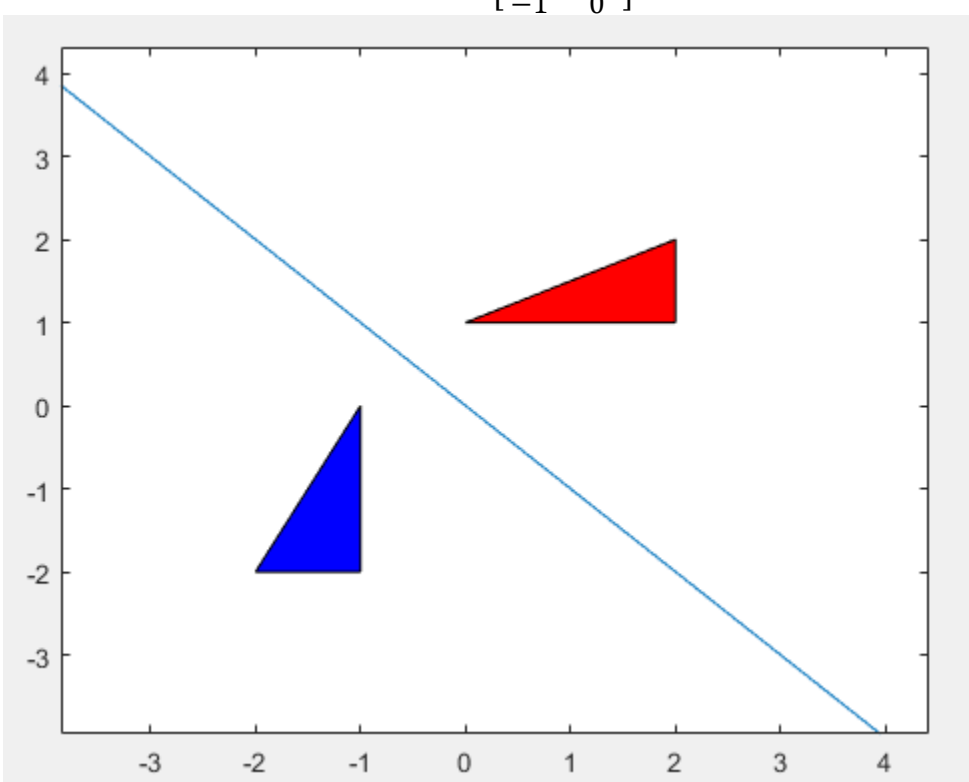
Oppgave 1

a)



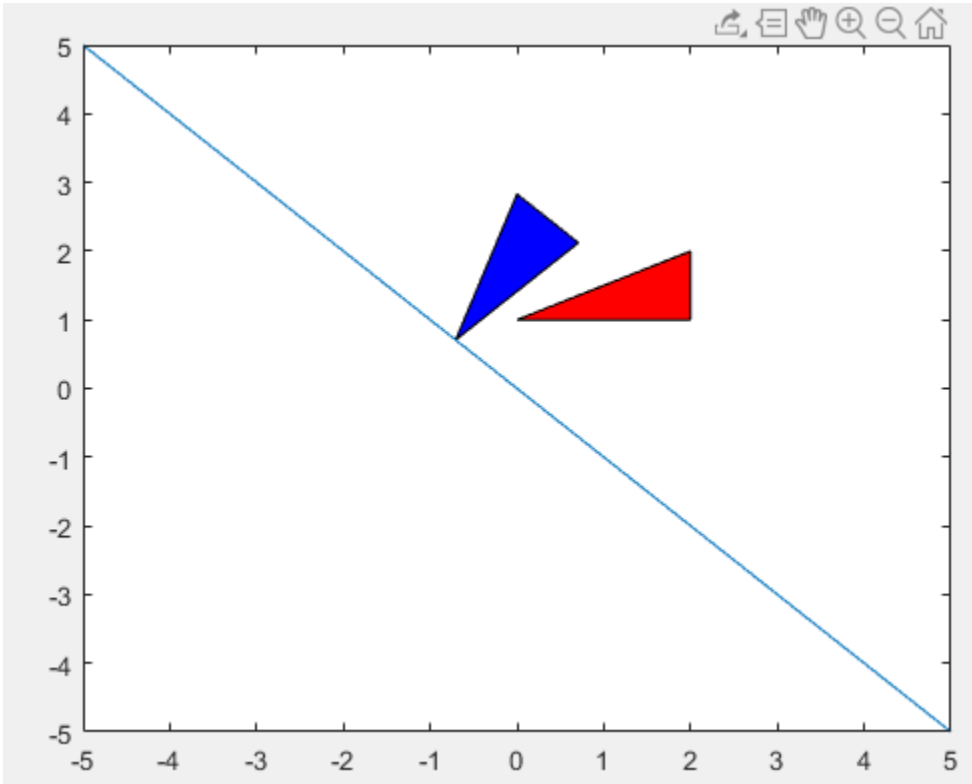


$$M = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$$

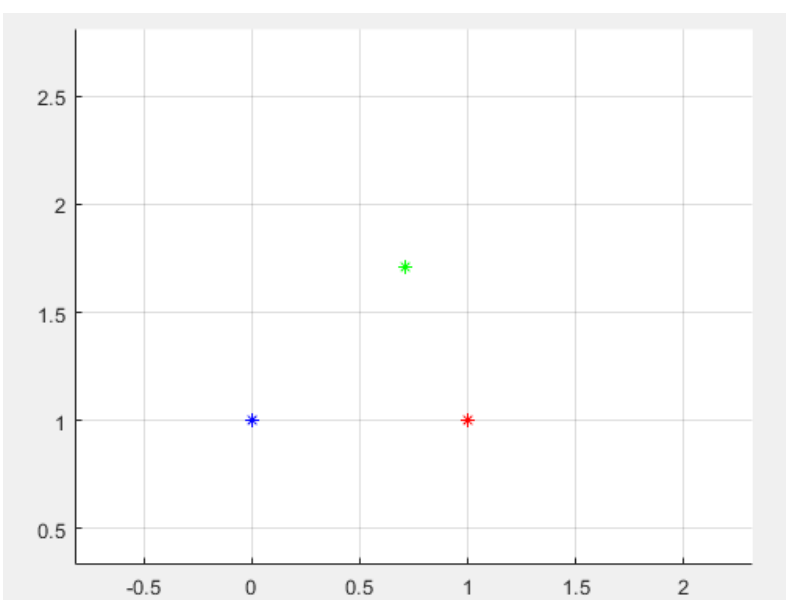


b)

$$M = \begin{bmatrix} \cos(45°) & -\sin(45°) \\ \sin(45°) & \cos(45°) \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{bmatrix}$$



c)



d)

$$T(\vec{a}) = \left(\begin{bmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{bmatrix} \times \left(\begin{bmatrix} a_x \\ a_y \end{bmatrix} - \begin{bmatrix} p_x \\ p_y \end{bmatrix} \right) \right) + \begin{bmatrix} p_x \\ p_y \end{bmatrix}$$

$$Linear\ Transformation \Leftrightarrow\ T(\vec{a} + \vec{b}) = T(\vec{a}) + T(\vec{b}) \wedge cT(\vec{a}) = T(c\vec{a})$$

$$T(\vec{a} + \vec{b}) = \begin{bmatrix} \frac{\sqrt{2}}{2}(a_x + b_x - p_x) - \frac{\sqrt{2}}{2}(a_y + b_x - p_y) \\ \frac{\sqrt{2}}{2}(a_x + b_x - p_x) + \frac{\sqrt{2}}{2}(a_y + b_x - p_y) \end{bmatrix} + \begin{bmatrix} p_x \\ p_y \end{bmatrix}$$

$$= \begin{bmatrix} \frac{\sqrt{2}}{2}(a_x + b_x - p_x) - \frac{\sqrt{2}}{2}(a_y + b_x - p_y) + p_x \\ \frac{\sqrt{2}}{2}(a_x + b_x - p_x) + \frac{\sqrt{2}}{2}(a_y + b_x - p_y) + p_y \end{bmatrix}$$

$$T(\vec{a}) + T(\vec{b}) = \begin{bmatrix} \frac{\sqrt{2}}{2}(a_x - p_x) - \frac{\sqrt{2}}{2}(a_y - p_y) \\ \frac{\sqrt{2}}{2}(a_x - p_x) + \frac{\sqrt{2}}{2}(a_y - p_y) \end{bmatrix} + \begin{bmatrix} p_x \\ p_y \end{bmatrix}$$

$$+ \begin{bmatrix} \frac{\sqrt{2}}{2}(b_x - p_x) - \frac{\sqrt{2}}{2}(b_y - p_y) \\ \frac{\sqrt{2}}{2}(b_x - p_x) + \frac{\sqrt{2}}{2}(b_y - p_y) \end{bmatrix} + \begin{bmatrix} p_x \\ p_y \end{bmatrix}$$

$$= \begin{bmatrix} \frac{\sqrt{2}}{2}(a_x - p_x) - \frac{\sqrt{2}}{2}(a_y - p_y) + p_x \\ \frac{\sqrt{2}}{2}(a_x - p_x) + \frac{\sqrt{2}}{2}(a_y - p_y) + p_y \end{bmatrix} + \begin{bmatrix} \frac{\sqrt{2}}{2}(b_x - p_x) - \frac{\sqrt{2}}{2}(b_y - p_y) + p_x \\ \frac{\sqrt{2}}{2}(b_x - p_x) + \frac{\sqrt{2}}{2}(b_y - p_y) + p_y \end{bmatrix}$$

$$T(\vec{a} + \vec{b}) \neq T(\vec{a}) + T(\vec{b}) \Rightarrow Ikke\ L.T.$$

e)

```
function ret = rotate(q,p,rad)
    rotationmatrix = [cos(rad), -sin(rad); sin(rad), cos(rad)];
    nyq = q - p;
    nyp = rotationmatrix*nyq;
    ret = nyp + p;
end
```

2

a)

$$\left| \begin{array}{ccc} + & - & + \\ \end{array} \begin{bmatrix} 2 & 2 & 0 \\ 2 & 1 & 1 \\ -1 & 0 & 3 \end{bmatrix} \right|$$

$$= 2\begin{vmatrix} 1 & 1 \\ 0 & 3 \end{vmatrix} - 2\begin{vmatrix} 2 & 1 \\ -1 & 3 \end{vmatrix} + 0\begin{vmatrix} 2 & 1 \\ -1 & 0 \end{vmatrix}$$

$$= 2(1\cdot 3 - 1\cdot 0) - 2(2\cdot 3 - 1\cdot -1) + 0$$

$$= 6 - 14 = -8$$

b)

$$\begin{array}{ccc} + & - & + \\ \end{array}$$

$$\begin{bmatrix} 2 & 1 & 1 \\ 2 & 2 & 0 \\ -1 & 0 & 3 \end{bmatrix}$$

$$2(2\cdot 0 - 0\cdot 0) - 1(2\cdot 3 - 0\cdot -1) + 1(2\cdot 0 - 2\cdot -1)$$

$$\underline{\underline{-6 + 2 = -4}}$$

c)

$$\pm \frac{1}{2} \begin{array}{c} \begin{array}{ccc} + & - & + \end{array} \\ \begin{vmatrix} 1 & 1 & 1 \\ -1 & 2 & 1 \\ 0 & 4 & 1 \end{vmatrix} \end{array}$$

$$= \pm \frac{1}{2}\left[1(2\cdot 1 - 1\cdot 4) - 1(-1\cdot 1 - 1\cdot 0) + 1(-1\cdot 4 - 2\cdot 0)\right]$$

$$= \pm \frac{1}{2}(-5)$$

$$= \underline{\underline{2.5}}$$

d)

Siden vi har allerede funnet determinanten, og den ikke er 0, har vi vist at det fins en invers av A.

3)

a)

$$\begin{bmatrix} 1 & 1/2 \\ 1/2 & 1/3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1/2 & 1/3 & 1/4 & 1/5 \\ 1/2 & 1/3 & 1/4 & 1/5 & 1/6 \\ 1/3 & 1/4 & 1/5 & 1/6 & 1/7 \\ 1/4 & 1/5 & 1/6 & 1/7 & 1/8 \\ 1/5 & 1/6 & 1/7 & 1/8 & 1/9 \end{bmatrix}$$

b)

$$\det(H) = 1 * \frac{1}{3} - \frac{1}{2} * \frac{1}{2} = \frac{1}{3} - \frac{1}{4} = \frac{1}{12}$$

$$adj(H) = \begin{bmatrix} \frac{1}{3} & -\frac{1}{2} \\ -\frac{1}{2} & 1 \end{bmatrix}$$

$$inv(H) = \frac{1}{\det(H)} adj(H) = \begin{bmatrix} 4 & -6 \\ -6 & 12 \end{bmatrix}$$

c)

```
1.0e+12 *

 0.0000   -0.0000    0.0000   -0.0000    0.0000   -0.0000    0.0000   -0.0000    0.0000   -0.0000
-0.0000    0.0000   -0.0000    0.0000   -0.0002    0.0005   -0.0008    0.0008   -0.0004    0.0001
 0.0000   -0.0000    0.0001   -0.0010    0.0043   -0.0112    0.0178   -0.0166    0.0085   -0.0018
-0.0000    0.0000   -0.0010    0.0082   -0.0379    0.1010   -0.1616    0.1529   -0.0788    0.0171
 0.0000   -0.0002    0.0043   -0.0379    0.1767   -0.4772    0.7712   -0.7358    0.3820   -0.0832
-0.0000    0.0005   -0.0112    0.1010   -0.4772    1.3014   -2.1207    2.0375   -1.0642    0.2330
 0.0000   -0.0008    0.0178   -0.1616    0.7712   -2.1207    3.4802   -3.3635    1.7658   -0.3883
-0.0000    0.0008   -0.0166    0.1529   -0.7358    2.0375   -3.3635    3.2674   -1.7230    0.3804
 0.0000   -0.0004    0.0085   -0.0788    0.3820   -1.0642    1.7658   -1.7230    0.9122   -0.2021
-0.0000    0.0001   -0.0018    0.0171   -0.0832    0.2330   -0.3883    0.3804   -0.2021    0.0449

  2.164452890406430e-53
```